Revisiting Communication in Federated Learning: Parameter Vs. Knowledge-Based Strategies

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Abstract

Federated Learning (FL) enables collaborative training without centralizing sensitive medical data, but conventional weight exchange incurs high communication costs and privacy risks. This work compares parameter-based and knowledge-based strategies for federated fall detection using three public datasets. While unweighted FedAvg achieves the strongest performance (0.9692 global PR-AUC), no-weight alternatives maintain competitive local results. Notably, soft-label communication reduces the communication payload from 49,665 parameters to only two values per round, offering the most efficient trade-off. These findings demonstrate the potential of lightweight knowledge-sharing for resource-constrained wearable healthcare systems.

1 Introduction

Federated Learning (FL) enables collaborative model training without centralizing sensitive medical data, making it particularly attractive for wearable healthcare applications [1, 2]. However, most FL systems rely on repeated exchanges of model parameters through Federated Averaging (FedAvg) [3], which introduces substantial communication overhead and remains vulnerable to information leakage through gradients and model updates [4, 5].

Recent research has explored communication-efficient alternatives to parameter sharing, including knowledge-distillation-based approaches, soft-label communication, and prototype-based aggregation [6, 7, 8]. However, their effectiveness in heterogeneous wearable sensing scenarios remains insufficiently understood.

This paper compares four communication strategies for federated fall detection: Federated Averaging (FedAvg), FedAvg with Knowledge Distillation (KD), soft-label communication, and prototype-based communication. The objective is to quantify the trade-off between predictive performance and communication efficiency.

2 Methodology

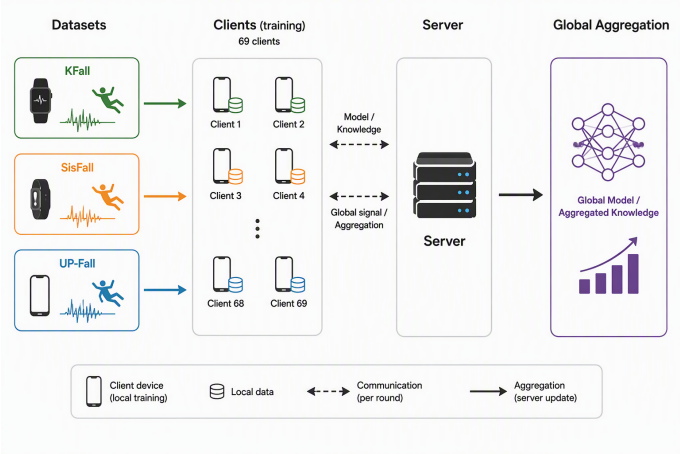


Figure 1: Federated learning framework adopted in this work. Soft-label communication exchanges aggregated Fall and Non-Fall probabilities, whereas prototype communication exchanges class-wise embeddings extracted from the 64-dimensional penultimate layer.

Experiments were conducted using the KFall, SisFall, and UP-Fall datasets [9, 10, 11]. Data were partitioned subject-wise into 69 federated clients, creating a heterogeneous non-IID setting with complete separation between training, validation, and test subjects. Figure 1 illustrates the federated learning setup adopted in this work. Depending on the evaluated strategy, clients exchange model parameters or compact knowledge representations.

Four approaches were evaluated: (i) Federated Averaging (FedAvg), (ii) FedAvg augmented with Knowledge Distillation (KD), (iii) Soft-Label Communication, and (iv) Prototype Communication [3, 12, 6, 7, 8, 13]. FedAvg and KD exchange full model parameters during each communication round. Soft-Label Communication exchanges two aggregated class probabilities (Fall and Non-Fall), whereas Prototype Communication exchanges class-wise latent embeddings extracted from the penultimate 64-dimensional representation layer.

All approaches used the same local classifier to ensure a fair comparison. The model consisted of a multilayer perceptron (MLP) with architecture 32–256–128–64–1 and dropout regularization (0.3). Training was performed for 25 communication rounds and 100 local epochs per round. Results are reported as averages over 10 independent runs. Local PR-AUC was used as the primary evaluation metric, while global PR-AUC was used to assess cross-client generalization.

Table 1: Experimental setup.

Component	Configuration
Datasets	KFall, SisFall, UP-Fall
Training Clients	69
Communication Rounds	25
Local Epochs	100
Model	MLP (32-256-128-64-1)
Runs	10 seeds

3 Results

Table 2: Best-performing configuration for each communication strategy.

Strategy	Local PR-AUC	Global PR-AUC	Payload
FedAvg	**0.9153**	**0.9692**	49,665
KD	0.9143	0.9573	49,665
Soft Labels	0.8169	**0.7365**	**2**
Prototypes	**0.8186**	0.7005	≈ 131

Table 2 summarizes the best-performing configuration obtained for each communication strategy.

Among all evaluated methods, unweighted FedAvg achieved the strongest performance, reaching a local PR-AUC of 0.9153 and a global PR-AUC of 0.9692. Interestingly, unweighted aggregation consistently outperformed weighted aggregation, suggesting that equal client contribution is beneficial under the heterogeneous data distributions created from KFall, SisFall and UP-Fall.

Knowledge Distillation produced performance very close to that of FedAvg, achieving a local PR-AUC of 0.9143 and a global PR-AUC of 0.9573. However, no measurable improvement was observed despite the additional teacher-student optimization objective, indicating that in-

troducing KD alone is insufficient when full model parameters are already exchanged. This observation is consistent with previous studies reporting that knowledge distillation mainly benefits communication-efficient settings rather than conventional parameter-sharing FL [6, 7].

The no-weight approaches exhibited substantially lower global performance. Prototype communication achieved the highest local PR-AUC among no-weight methods (0.8186), demonstrating that useful client models can still be learned without weight sharing, aligning with recent prototype-based FL approaches designed to improve learning under heterogeneous client distributions [8, 13]. However, the improvement over soft-label communication was marginal (0.8186 vs. 0.8169), despite requiring approximately 65 times more transmitted values.

Soft-label communication achieved a slightly lower local PR-AUC but obtained the best global performance among no-weight strategies (0.7365 vs. 0.7005). As illustrated in Figure 2, it also reduced the communication payload from 49,665 transmitted parameters to only two scalar values per communication round, corresponding to approximately 99.996% less transmitted information.

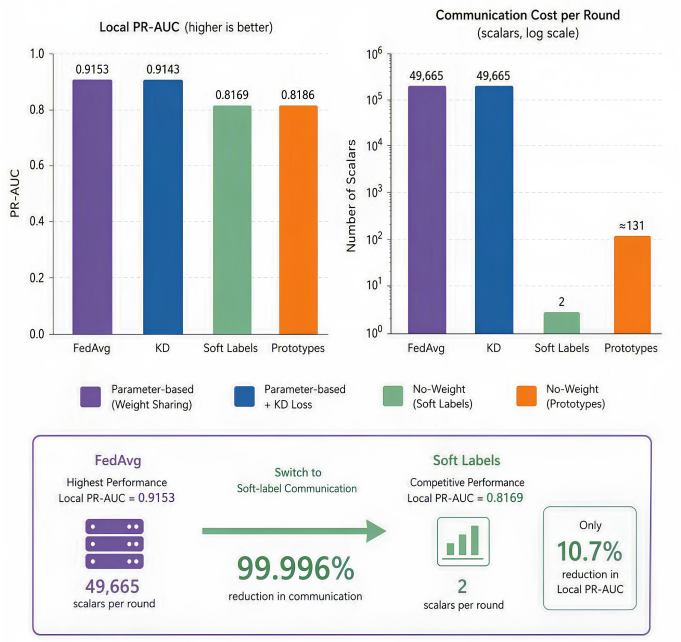


Figure 2: Communication-performance trade-off between predictive performance and communication cost.

As shown in Figure 2, soft-label communication provides the most favorable trade-off between communication efficiency and predictive performance. Although FedAvg achieves the highest local PR-AUC, soft labels reduce the communication payload from 49,665 transmitted parameters to only two scalar values per communication round.

4 Discussion

The results highlight a clear distinction between predictive performance and communication efficiency. As expected, FedAvg achieved the strongest global and local performance, suggesting that full parameter exchange remains the most effective approach for aggregating knowledge across heterogeneous clients.

FedAvg's superior performance stems from the richer information content in full model parameters. Unlike soft labels or prototypes, complete parameter updates preserve detailed optimization information from each client, allowing the aggregation process to better capture heterogeneous data distributions.

Knowledge Distillation produced performance comparable to FedAvg but offered no measurable gains. This suggests that modifying the local optimization objective alone is insufficient when the communication mechanism remains unchanged.

Among the no-weight approaches, soft-label communication achieved the strongest overall trade-off, obtaining a local PR-AUC of 0.8169 while requiring only two transmitted values per communication round. Prototype communication achieved a comparable local PR-AUC (0.8186), but required a substantially larger communication payload (approximately 131 transmitted values). Although prototype aggregation has shown promise in heterogeneous federated environments [8, 13], the additional information communicated did not translate into better global performance in this fall-detection scenario.

These findings are particularly relevant for wearable and mobile healthcare systems, where communication bandwidth, battery consumption, and privacy constraints often limit the practicality of repeated model-parameter exchange.

5 Conclusion

This work compared parameter-based and knowledge-based communication strategies for federated fall detection using three public wearable datasets. Although FedAvg remained the strongest approach in terms of predictive performance, communication-efficient strategies achieved competitive local performance with drastically reduced communication requirements.

In particular, soft-label communication reduced the transmitted payload from 49,665 parameters to only two scalar values per communication round (99.996% reduction), while maintaining a local PR-AUC of 0.8169. These results suggest that lightweight knowledge-sharing mechanisms constitute practical alternatives for communication-constrained federated healthcare applications.

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